MIL-M-26788C (USAF)
26 February 1974
SUPERSEDING
MIL-M-26788B (USAF)
1 November 1969

MILITARY SPECIFICATION

MANUALS, TECHNICAL OPERATION AND OPERATOR MAINTENANCE INSTRUCTIONS (For Automotive Equipment)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the minimum requirements for the preparation of technical manual(s) for operation and operator maintenance of automotive equipment, i.e., trucks, tractors, truck tractors, semitrailers, forklifts, etc.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-M-38784	Manual, Technical: General Requirements for Preparation of
MIL-P-38790	Printing Production of Technical Manuals: General Requirements for

(Copies of documents required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preparation. The general manner of preparation shall be in accordance with MIL-M-38784 and MIL-P-38790.

3.1.1 Contents. The manual shall include illustrations and instructions for the operation and operator maintenance of the equipment and its accessories. The instructions shall include an introduction which will reference, when applicable, other manuals (commercial and Air Force Publications) that pertain to operation and operator maintenance. Sections on description, operation, and inspections, maintenance and lubrication for which the operator is responsible, will be furnished. Any special instructions for the operator will be included under the appropriate section. Warranty provisions will not be included.

3.2 Arrangement. The manual shall be arranged as follows:

Front Matter

Title Page
List of Effective Pages
Table of Contents
List of Illustrations
List of Tables
Introduction

Section I, Description
Section II, Operation
Section III, Operator insepction, maintenance and lubrication.

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3.2.1 Front matter. The title page, list of effective pages, table of contents, list of illustrations, list of tables, and introduction shall be prepared in accordance with MIL-M-38784.

3.2.2 Section I, Description. This section shall include a concise, complete description of the equipment, purpose for which equipment is to be used, general characteristics, and performance. This shall be followed by a brief, readily understood description of component assemblies which includes the complete equipment, presented in a manner indicating relationship and function of components.

3.2.3 Section II, Operation. This section shall include operating principles, and instructions for operating equipment. Illustrations, sketches, wiring and system diagrams, shall be used when necessary for a complete understanding of the subject. Complete operating instructions shall include the following, with additions or deletions as necessary.

a. Safety precautions.

b. Purpose and use of operating controls.

c. Purpose and use of all indicating instruments.

d. Equipment start-up procedures.

e. Equipment normal operating procedures.

f. Accessory equipment operating procedures, including installed emergency equipment.

g. Equipment shut down procedures, normal and emergency.

3.2.4 Section III, Operator inspection, maintenance and lubrication. This section shall include, as applicable, instructions for operator fault isolation of equipment malfunctions, all periodic inspections, operators servicing instructions (such as: servicing of coolant, fuel and lubricating systems, battery, stearing reservior, transmission etc).

3.2.4.1 Equipment fault isolation and correction. The most common malfunctions which are likely to occur during equipment operation, a statement of probable cause(s), and a satisfactory operator remedy for each cause shall be included in tabular format as shown in the example, Figure 1. This information should be provided in the sequence of probability of occurence. Remedial action will include referral to maintenance shop for repair when such repairs are beyond the scope of operator maintenance.

3.2.4.2 Operator inspection. All periodic inspections to be performed by the operator shall be included in a table, as shown in the example, Figure 2. Inspection periods shall be expressed in miles, hours of operation, or maximum period of time that may elapse between inspections if the prescribed mileage or hours of operation have not been reached.

3.2.4.3 Operator maintenance and servicing. Operator maintenance and servicing instructions shall be provided. Components which require lubrication by the operator shall be identified and illustrated per Figure 3. The type, amount, and specification number of lubricant will be included. This information will cover temperature ranges from minus 65 degrees Fahrenheit to plus 160 degrees Fahrenheit.

4. QUALITY ASSURANCE PROVISIONS

4.1 Inspection and approval of the manual shall be in accordance with MIL-M-38784 and MIL-P-38790.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with MIL-M-38784 and MIL-P-38790.

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6. NOTES

6.1 Intended use. Technical manuals prepared in accordance with this specification are for use in operating automotive equipment and for performing operator maintenance.

Custodian:	Preparing Activity
Air Force - 16	Air Force - 16
Review Activities:	Project No.
Air Force - 01, 10, 70	TMSS F384

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TROUBLE	PROBABLE CAUSE	CHECKOUT PROCEDURE AND REMEDIAL ACTION
Engine will not crank.	1. Battery discharged. 2. Battery terminal loose or corroded. 3. Transmission selector valve not in neutral position.	1. Refer to maintenance shop for charge or replacement of battery. 2. Tighten or clean terminals. 3. Place selector valve in neutral.
Engine cranks but will not start.	1. Fuel tank empty. 2. Emergency fuel shut-off switch in OFF position.	1. Check fuel gauge and fill tank, if required. 2. Place in normal run position
Engine overheats.	1. Insufficient coolant 2. Broken fanbelt.	1. Add coolant. 2. Replace fanbelt.
Vehicle will not move when shifted into gear with engine running.	1. Parking brake not released. 2. Insufficient air pressure to operate shift controls. 3. Transmission fluid level too low.	1. Release parking brake. 2. Assure that all drain valves are closed and allow time for air system to build up pressure. If pressure does not build up, trouble shoot the air system. 3. Fill to proper level.

Figure 1. Troubleshooting Chart (Example)

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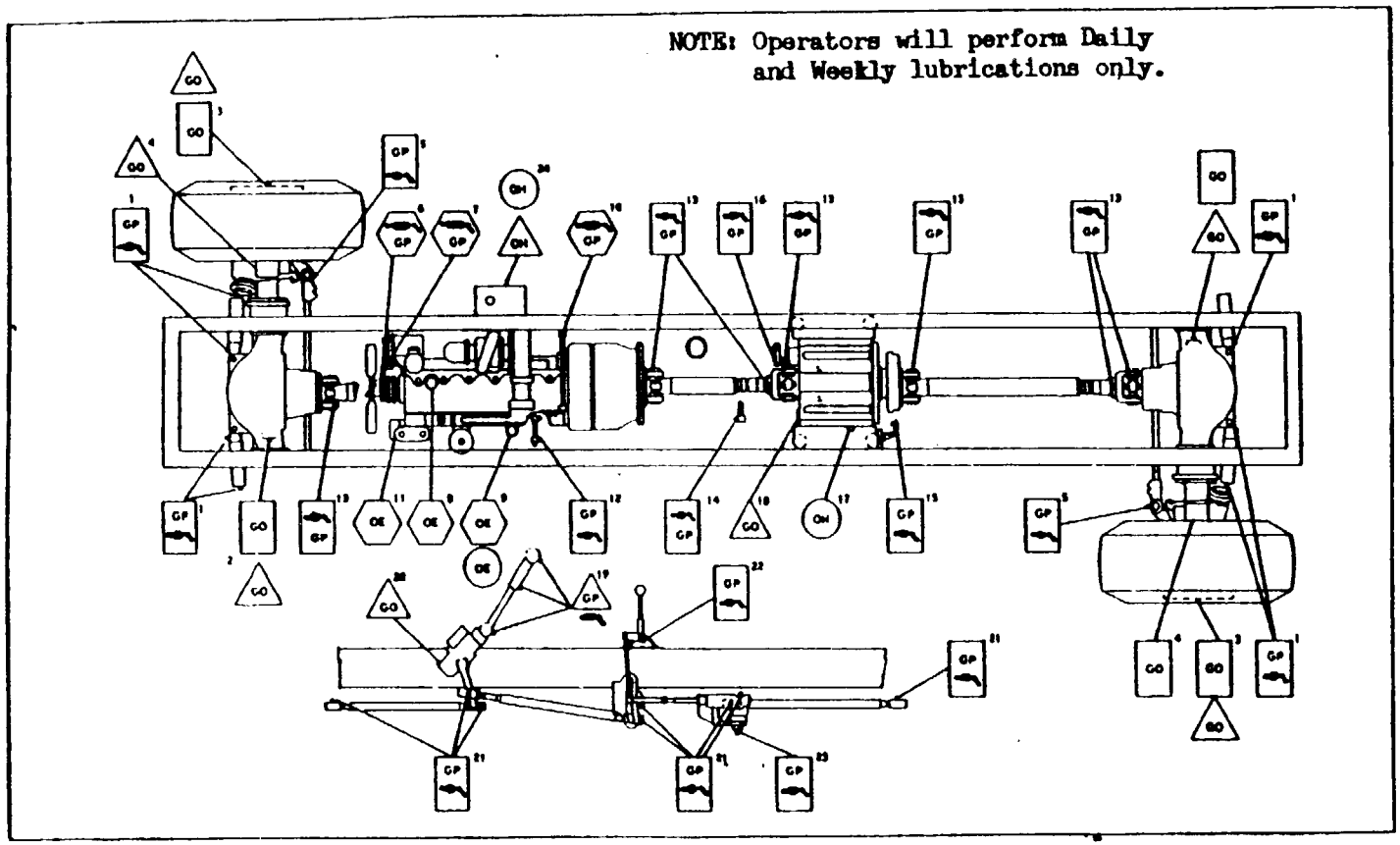
OPERATOR INSPECTION CHART

(Check daily when vehicle is used)

AIR SYSTEM Empty air tank to remove condensation. Check windshield wiper operation.
BATTERY Check water
BRAKES - SERVICE Check air lines for leaks. Check brakes for proper operation. Check brake lines for leaks.
BRAKES - PARKING Check for proper operation.
ENGINE Check oil level, add oil if necessary. Visually inspect for oil, water or fuel leaks, tighten connections as required. Check pressure and temperature gauge reading while operating vehicle. Listen for unusual engine noises.

Figure 2. Operator Inspection Chart (Example)

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1. Steering cylinder
2. Differential housing (see note 3)
3. Planetary housing (see note 4)
4. Universal joints and trunnion bearings (see note 5)
5. Tie rods
6. Fan pulley
7. Tachometer drive
8. Engine breather (see note 6)
9. Engine crankcase (see note 7)
10. Accelerator linkage
11. Lube oil cooler (see note 8)
12. Throttle linkage
13. Drive line bearings
14. Transmission shift rod
15. Parking brake bell crank
16. Speedometer adapter
17. Transmission sight gage (see note 9)
18. Transmission strainer (see note 10)
19. Steering column universal connector
20. Steering gear (see note 11)
21. Drag links
22. Steering shift lever
23. Steering bell crank
24. Hydraulic reservoir (see note 12)

Figure 3. Lubrication Chart (Sheet 1 of 2)

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LUBRICANT KEY				
INTERVAL	SYMBOLS AND TYPES OF LUBRICANTS	TEMPERATURE RANGES AND LUBRICANT SPECIFICATION		
		32°F (0°C) AND ABOVE	0°F (-18°C) TO 32°F (0°C)	0°F (-18°C) TO -65°F (-54°C)
DAILY ○	OE LUBRICATING OIL, ENGINE	OE-30 MIL SPEC MIL-L-45199	OE-10 MIL SPEC MIL-L-45199	MIL SPEC MIL-L-45199 (See Note 13)
MONTHLY 100 HRS ⬡	HYDRAULIC STEERING OH OIL, HYDRAULIC	MIL SPEC MIL-L-2104 10W-30	MIL SPEC MIL-L-2104 10W-30	MIL SPEC MIL-L-2104 10W-30
QUARTERLY 300 HRS □	TRANSMISSION OH OIL, HYDRAULIC	TYPE C MIL-L-2104	TYPE C MIL-L-2104	TYPE C MIL-L-2104
ANNUALLY 1200 HRS △	GO OIL, GEAR	OE-90 MIL SPEC VV-L-765	OE-90 MIL SPEC VV-L-765	OE-90 MIL SPEC VV-L-765
	GP GREASE, GENERAL PURPOSE	MIL SPEC MIL-G-10924	MIL SPEC MIL-G-10924	MIL SPEC MIL-G-10924

APPLICATION SYMBOLS	
	OIL CAN
	HAND
	GREASE GUN
	BRUSH

LUBRICATION NOTES

1. Clean the lubrication fittings with dry cleaning solvent, Federal Specification P-D-680. Dry before applying lubricant.

2. Intervals specified are for normal operation. The maintenance officer may adjust the intervals to compensate for unusual climatic or operational conditions.

3. Check the lubricant level quarterly or 300 operating hours whichever is first. Drain the lubricant while the differential is heated annually or 1200 hours and flush. Refill with new lubricant.

4. Check the lubricant level quarterly or 300 hours. Drain the planetary housing annually or 1200 operating hours. Flush and refill with new lubricant.

5. Drain the universal joint housing annually or 1200 hours. Flush and refill with new lubricant.

6. Clean the engine breather each month or 100 hours. Oil with OE lubricant.

7. Check oil level daily. Drain the crankcase every month or 100 hours. Refill with new OE lubricant.

8. Drain each time the engine crankcase is drained

9. Check oil level daily.

10. Drain the transmission annually or 1200 hours. Clean the strainer.

11. Drain and flush annually or 1200 hours. Refill with new lubricant.

12. Check the hydraulic oil level daily. Drain and refill with new oil annually or 1200 hours.

13. Dilute engine lubricating oil with Kerosene, Military Specification VV-K-211, for temperatures below 0°F (maximum dilution 25%).

NOTE: Operators will perform Daily and Weekly lubrications only.

Figure 3. Lubrication Chart (Sheet 2 of 2)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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